

Chapter 5

NLP Use cases : But not limited to this

x

y

auto complete

not interested at



this time

translation

how are you?



क्या हाल है?

NER

Rudolph Smith bought 1000 shares of tesla Inc. in March 2020



Rudolph Smith bought 1000 shares of tesla Inc. in March 2020

Sentiment
Analysis

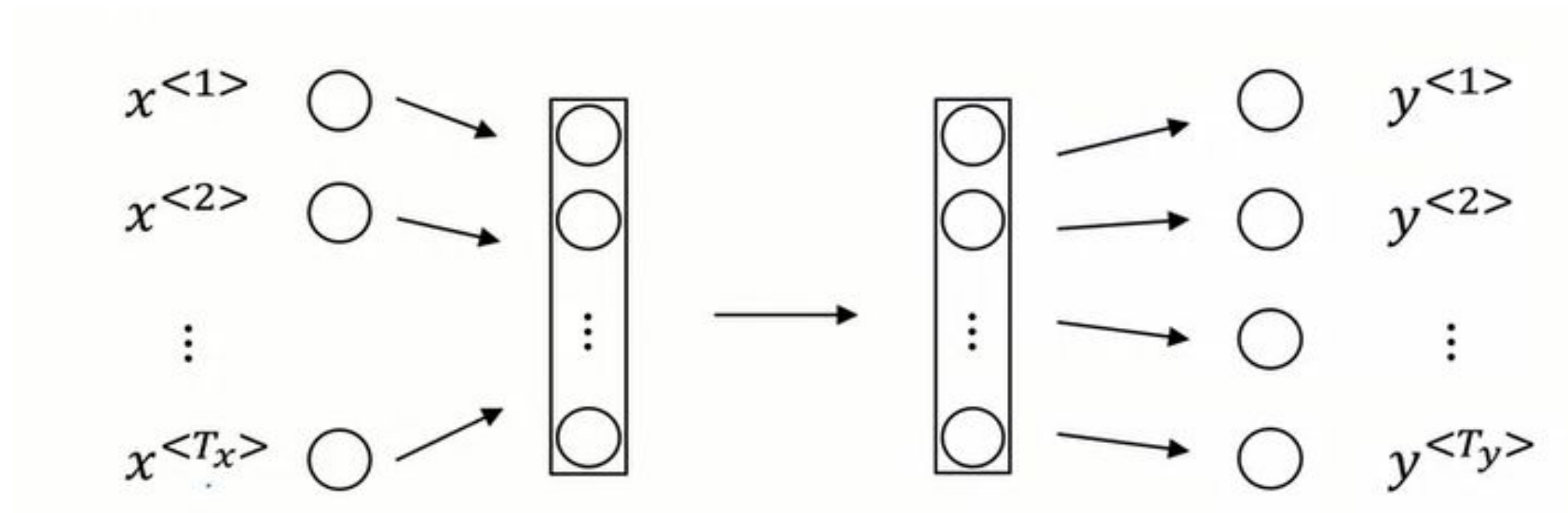
Not only the fan was expensive,
but it was broken when it
arrived.



Why not a standard network?

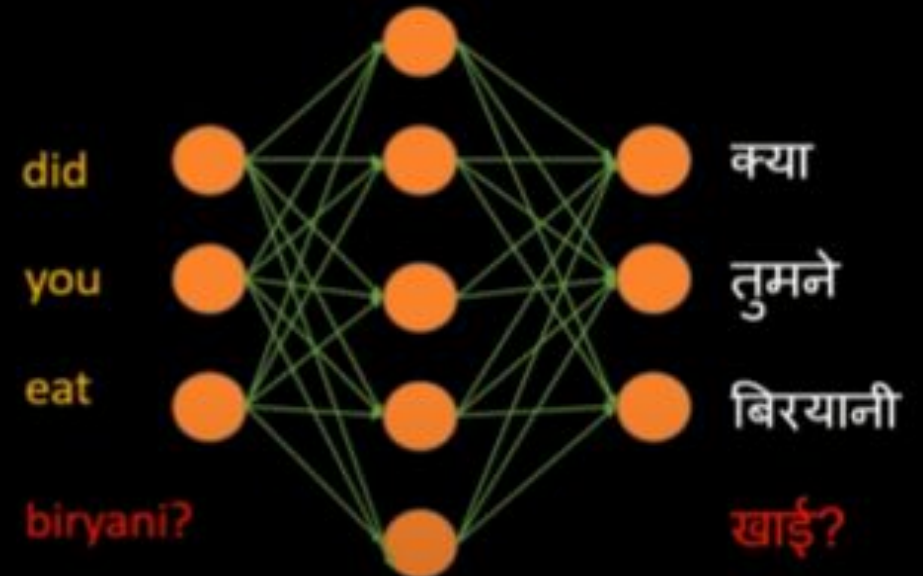
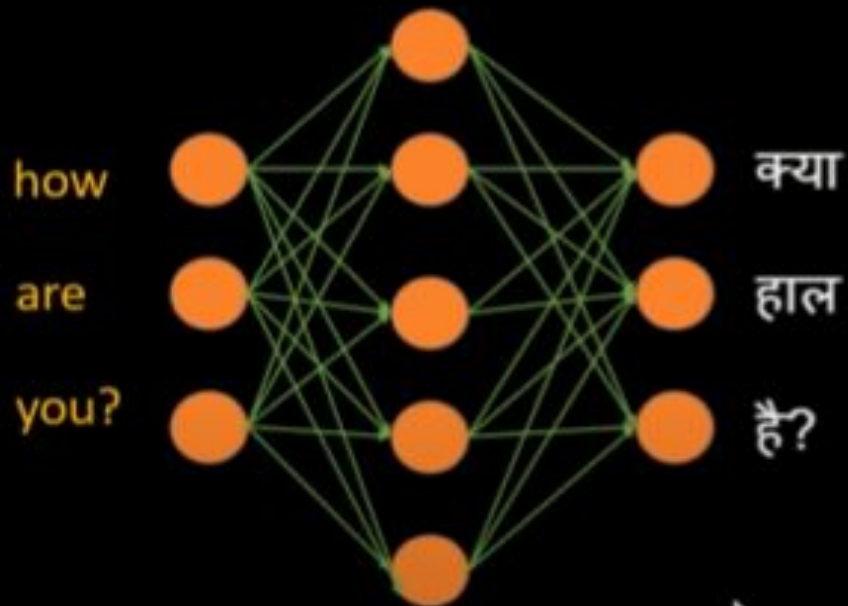
- Traditional feedforward neural networks do not share features across different positions of the network.
- In other words, these models assume that all inputs (and outputs) are independent of each other.
- This model would not work in sequence prediction since the previous inputs are inherently important in predicting the next output.
- For example, if you were predicting the next word in a stream of text, you would want to know at least a couple of words before the target word.

Standard network have certain issues



Why not a standard network?

Issue # 1: No fixed size of neurons in a layer



Why not a standard network?

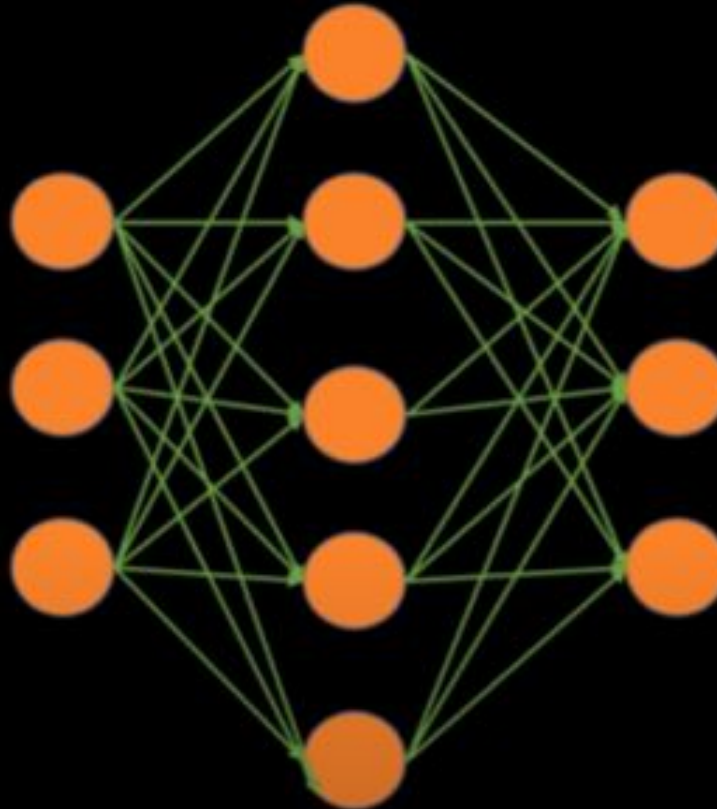
Issue # 2: Too much computation

25000 words in vocabulary

how $\rightarrow [0,0,0,\dots,1,0,0,\dots,0]$

are $\rightarrow [0,1,0,0,0,\dots,0,0,\dots,0]$

you? $\rightarrow [0,0,0,0,\dots,0,0,1,0,0]$



42000 words in vocabulary

$[0,0,0,\dots,1,0,0,\dots,0]$

$[0,0,0,\dots,1,0,0,\dots,0]$

$[0,0,0,\dots,1,0,0,\dots,0]$

क्या

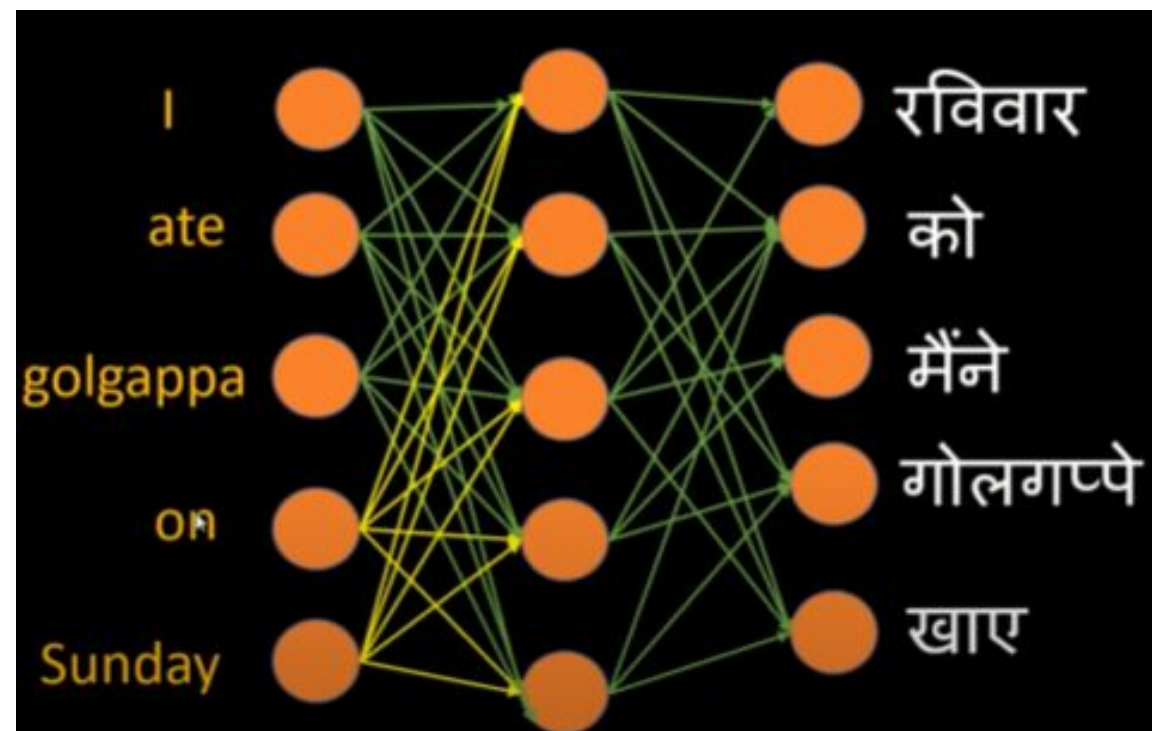
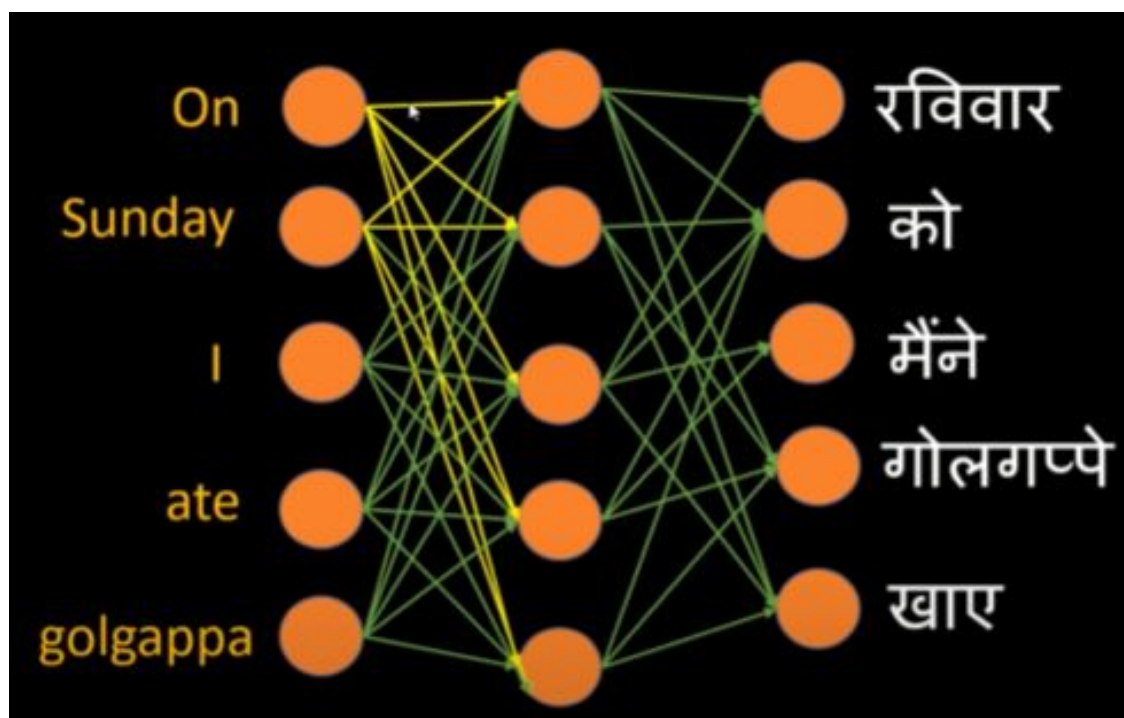
हाल

है?

On sunday I ate golgappa

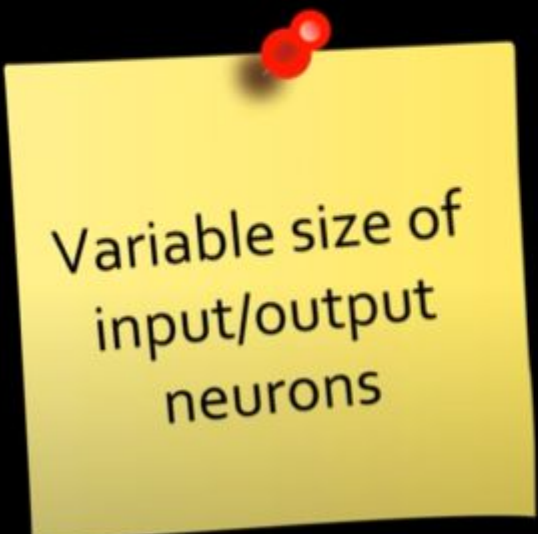
I ate golgappa on Sunday

रविवार को मैंने गोलगप्पे खाए



Issue # 3: Parameters are not shared

3 Issues using ANN for sequence problems



Variable size of
input/output
neurons

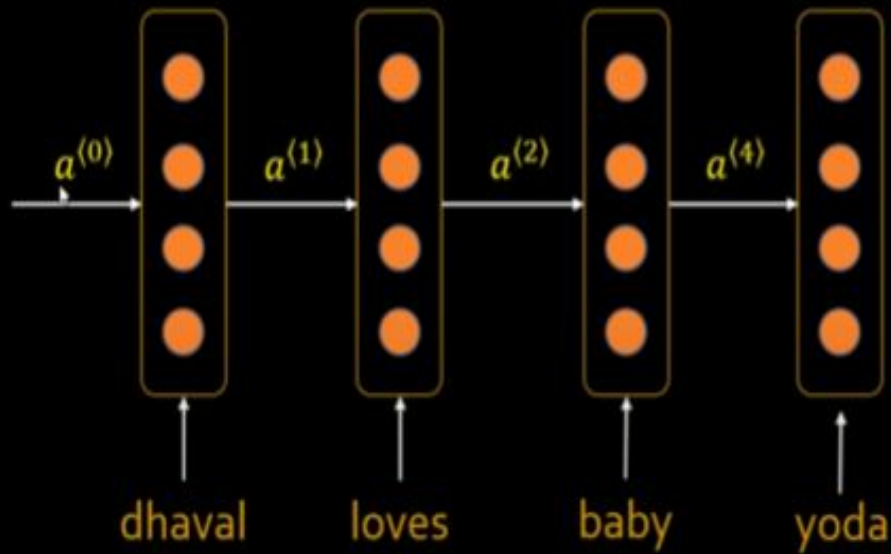


Too much
computation



No parameter
sharing

Named Entity Recognition

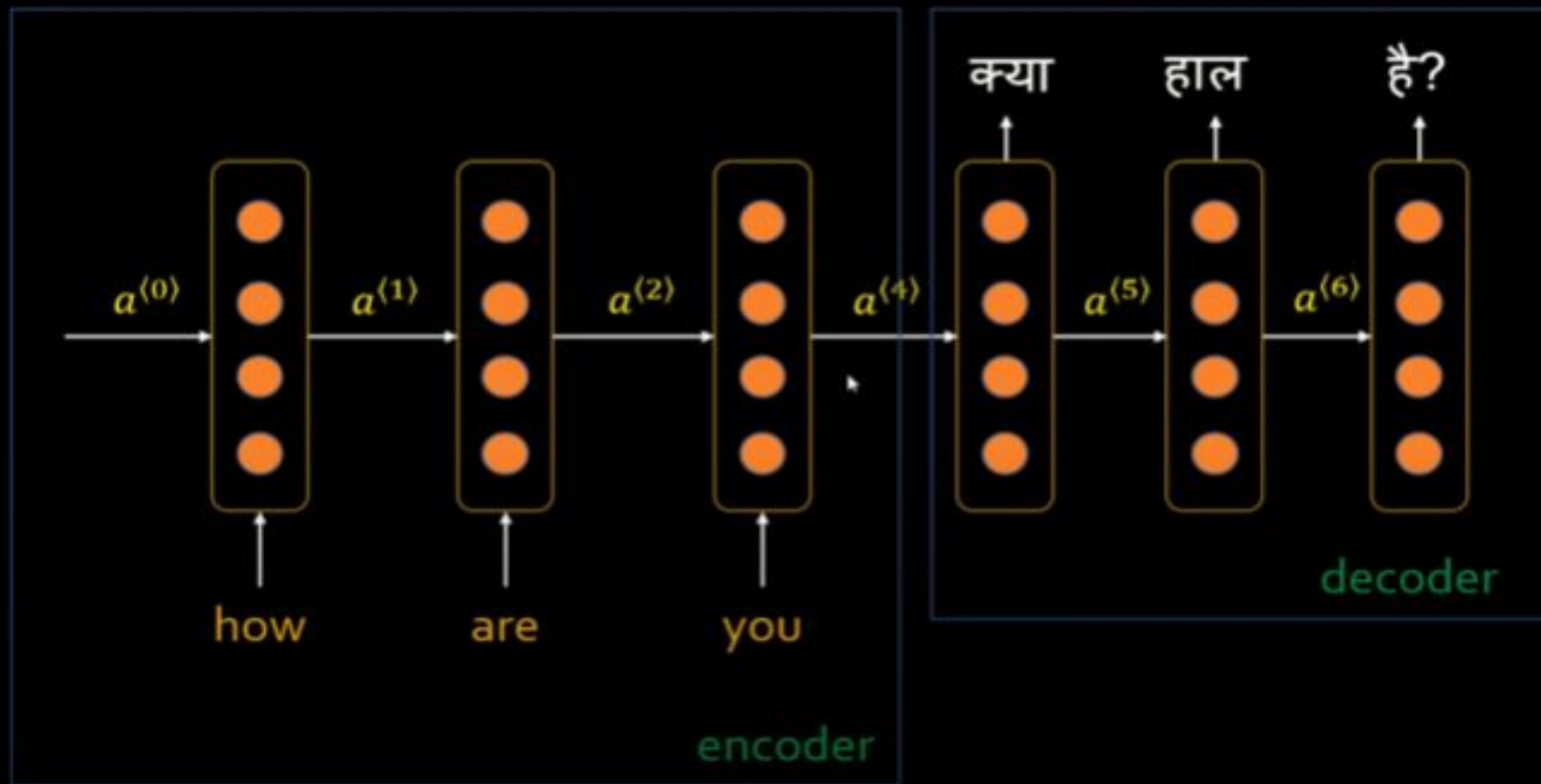


Generic Representation of RNN



Language translation

how are you? क्या हाल है?



LSTM

Today, due to my current job situation and family conditions, I need to take a loan.

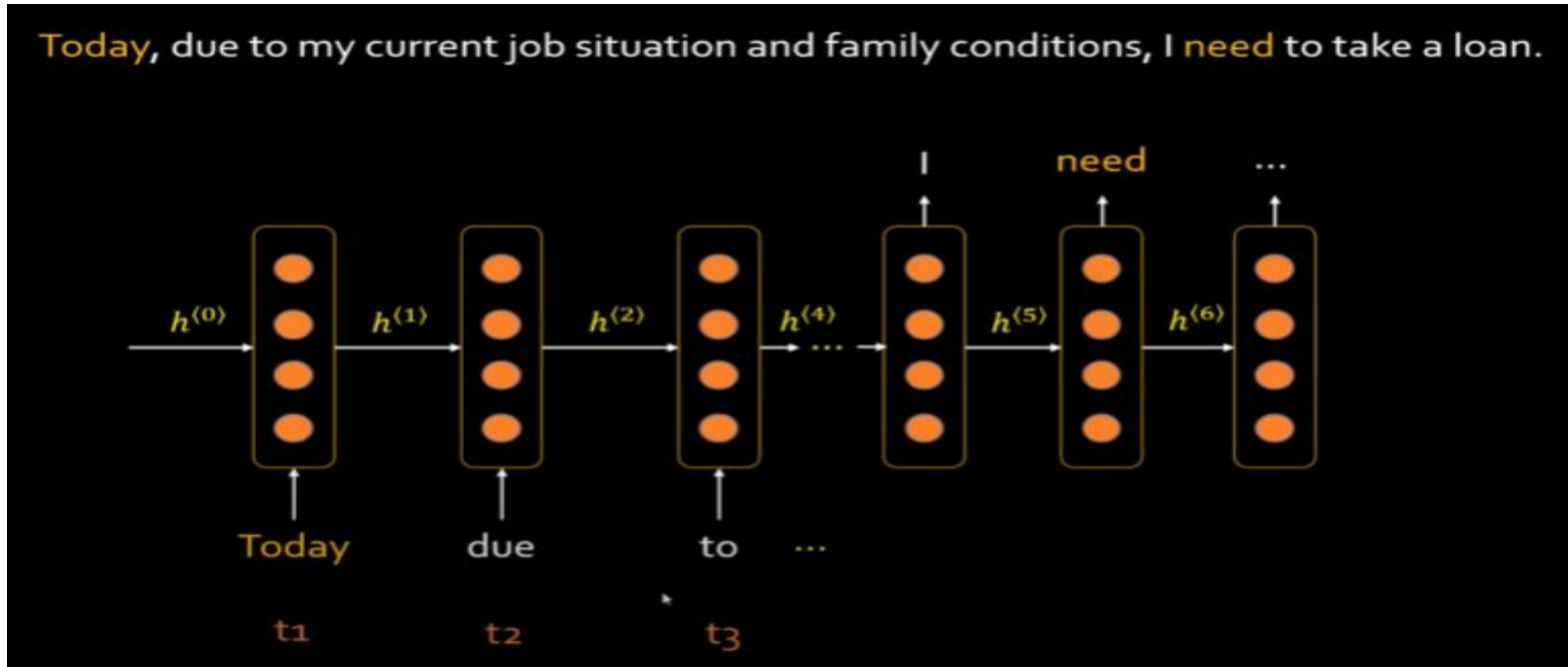
Last year, due to my current job situation and family conditions, I had to take a loan.

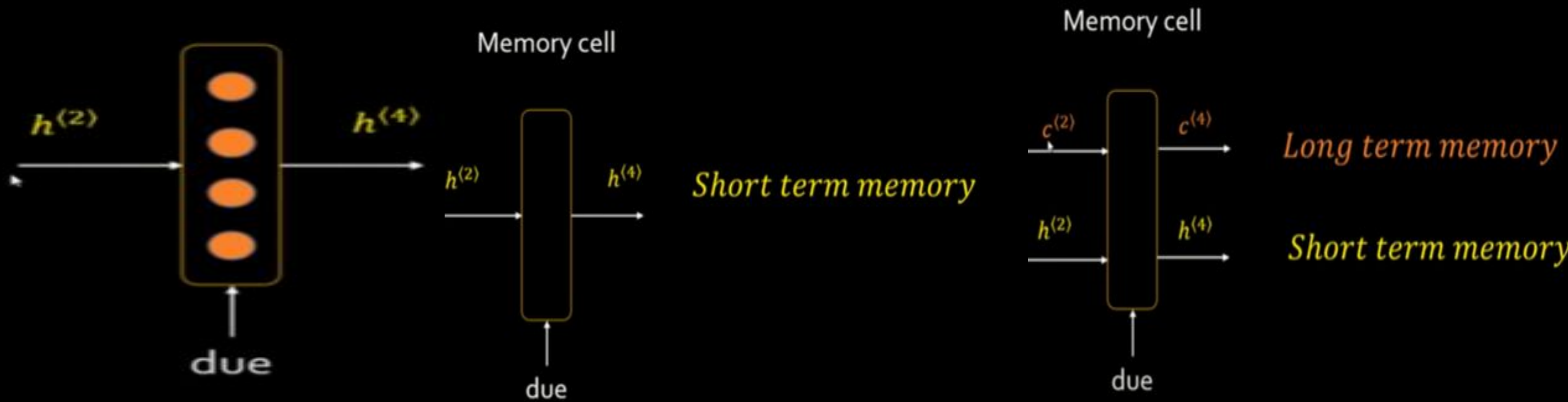
What happened with RNN?

Today, due to my current job situation and family conditions, I need to take a loan.



RNN Unrolled Over a time



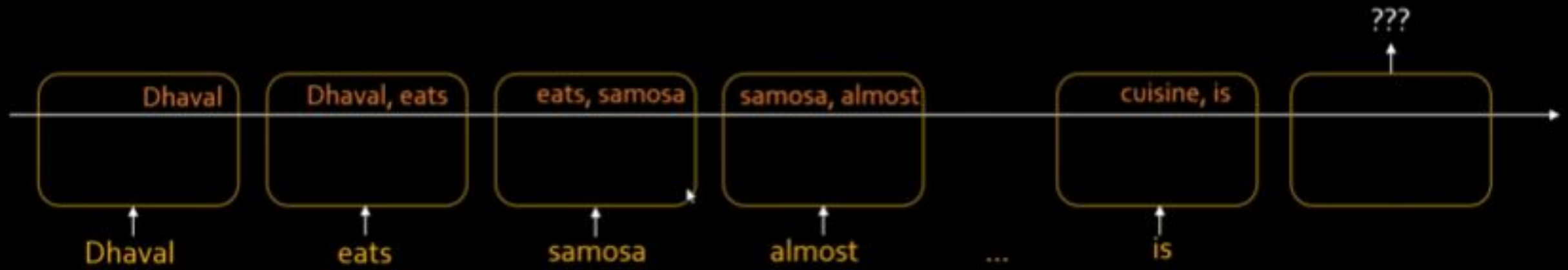


Self state c

Dhaval eats samosa almost everyday, it shouldn't be hard to guess that his favorite cuisine is Indian



Traditional RNN



Long term memory

Dhaval eats samosa almost everyday, it shouldn't be hard to guess that his favorite cuisine is Indian

